

9. Addition of two signals using Vivado IP Block based design

Aim

To design and simulate a Direct Digital Synthesizer (DDS) based system using Xilinx Vivado, where two sinusoidal signals (5 MHz and 10 MHz) generated using DDS Compiler IP cores are added together using an adder/subtractor block, and to observe the combined output waveform.

Software Required

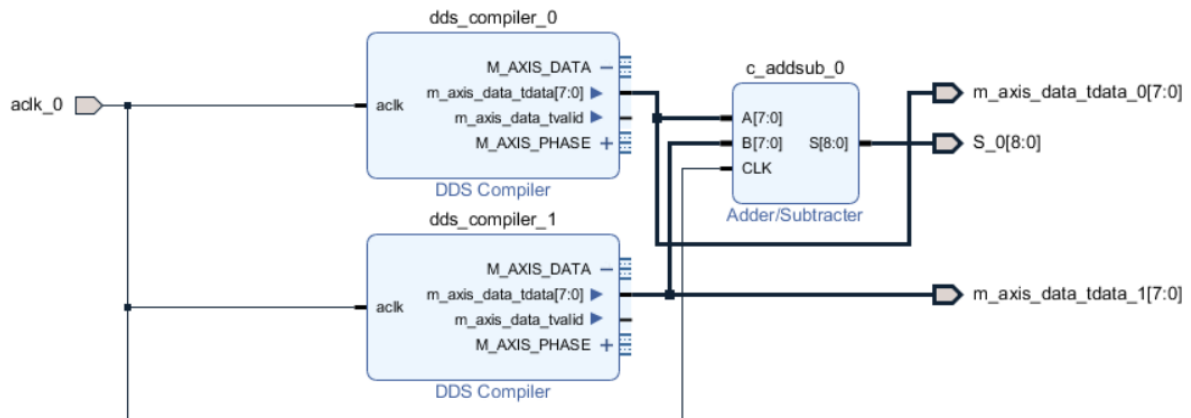
- Xilinx Vivado Design Suite
- Vivado IP Catalog
- Vivado Simulator (XSIM)

Procedure

1. Project Creation
 - Open Vivado and create a new RTL project.
 - Select the target FPGA device/board.
 - Choose *Do not specify sources at this time*.
2. Block Design Creation
 - Open IP Integrator → Create Block Design.
 - Add the following IPs:
 - Two DDS Compiler cores
 - One Adder/Subtractor core
 - One Clock source / external clock port
3. DDS Configuration
 - Configure dds_compiler_0 to generate a 5 MHz sine/cosine signal.
 - Configure dds_compiler_1 to generate a 10 MHz sine/cosine signal.
 - Use the same system clock for both DDS cores.
4. Connections
 - Connect both DDS clocks to the same aclk.
 - Connect m_axis_data_tdata outputs of both DDS cores to the inputs of the adder/subtractor block.
 - Connect the adder output to an external output port.
5. Wrapper and Simulation
 - Create HDL wrapper.
 - Write a simple testbench to generate clock.

- Run behavioral simulation.
- Observe individual DDS outputs and the summed output.

Block based design



Testbench program

```
`timescale 1ns / 1ps
```

```
module tb_adding_two_signals;
```

```
reg clk;
```

```
reg rst;
```

```
wire clk_out;
```

```
design_1_wrapper uut (
```

```
    .S_0(),
```

```
    .acik_0(clk),
```

```
    .m_axis_data_tdata_0(),
```

```
    .m_axis_data_tdata_1()
```

```
);
```

```
always #5 clk = ~clk;
```

```
initial begin
```

```
    clk = 0;
```

```
    rst = 1;
```

```

#20 rst = 0;

#100;

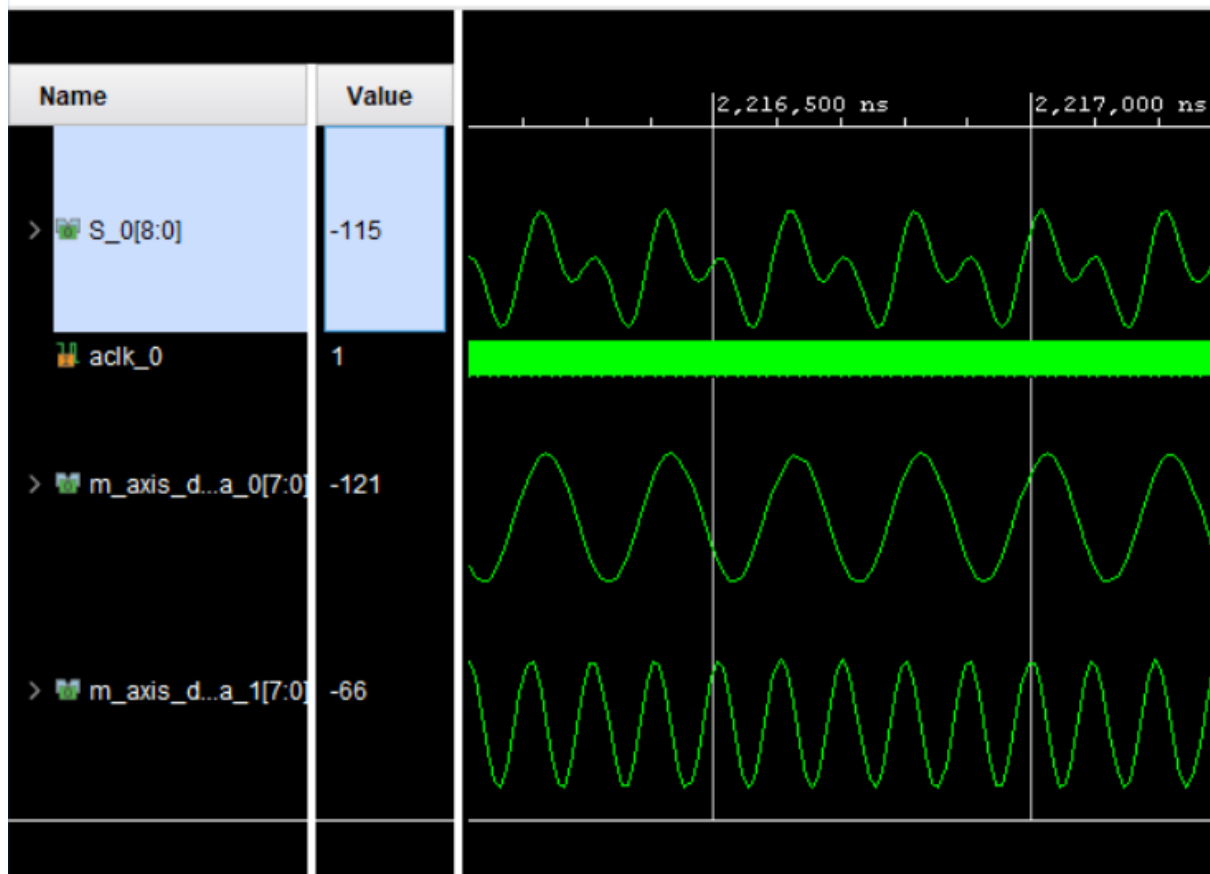
$stop;

end

endmodule

```

Output



Result

The DDS-based signal generator was successfully designed and simulated.

- dds_compiler_0 generated a 5 MHz sinusoidal waveform.
- dds_compiler_1 generated a 10 MHz sinusoidal waveform.
- The adder/subtractor block correctly added both signals, producing a combined waveform containing both frequency components.
- The simulation waveforms confirmed correct operation of the DDS cores and the signal addition process.